would become a matter of degree, and there would be no such thing as an absolutely analytic sentence. This would be such a departure from the classical Carnap-Ayer-et al. notion of analyticity that Quine feels that if *this* is what we mean to talk about, then it would be less misleading to introduce a different term altogether, say, *centrality*.

The second behavioral index is *being called "analytic."* In effect, some philosophers take the hallmark of analyticity to be that trained informants (say, Oxford dons) *call* the sentence analytic. Variants of this index are: that the sentence be deducible from the sentences in a finite list at the top of which someone who bears the ancestral of the graduate-student relation to Carnap has printed the words "Meaning Postulate"; that the sentence be obtainable from a theorem of logic by substituting synonyms for synonyms. The last of these variants looks promising, but Quine launches against it the question, "what is the criterion of synonymy?" One possible criterion might be that words W_1 and W_2 are synonymous if and only if the biconditional (x) (x is in the extension of $W_1 \equiv x$ is in the extension of W_2) is *analytic*; but this leads us right back in a circle. Another might be that words W_1 and W_2 are synonymous if and only if trained informants *call* them synonymous; but this is just our second index in a slightly revised form. A promising line is that words W_1 and W_2 are synonymous if and only if W_1 and W_2 are interchangeable (i.e., the words can be switched) *salva veritate* in all contexts of a suitable class. But Quine convincingly shows that this proposal too leads us around in a circle. Thus the second index reduces to this: a sentence is analytic if either it or some expression, or sequence of ordered pairs of expressions, or set of expressions, related to the sentence in certain specified ways, lies in a class to all the members of which trained informants apply a certain *noise*: either the *noise* ANALYTIC, or the *noise* MEANING POSTULATE, or the *noise* SYNONYMOUS. Ultimately, this proposal leaves "analytic," etc., *unexplicated noises*.

Although Quine does not discuss this explicitly, it is clear that taking the intersection of the two unsatisfactory behavioral indexes would be no more satisfactory; explicating the analyticity of a sentence as consisting in centrality *plus* being called ANALYTIC is just saying that the analytic sentences are a subclass of the central sentences without in any way telling us wherein the exceptionality of the subclass consists. In effect, Quine's conclusion is that analyticity is either centrality misconceived or it is nothing.

In spite of Quine's forceful argument, many philosophers have gone on abusing the notion of analyticity, often confusing it with a supposed highest degree of centrality. Confronted with Quine's alternatives, they have elected to identify analyticity with centrality, and to pay the price — the price of classifying such obviously synthetic-looking sentences as "space has three dimensions" as analytic, and the price of undertaking to maintain the view that there is, after all, such a thing as absolute unrevisability in science in spite of the impressive evidence to the contrary. But this line can be blasted by coupling Quine's argument with an important argument of Reichenbach's.

In his book *The Theory of Relativity and A Priori Knowledge*, Reichenbach showed that there exists a set of principles (see p. 31) each of which Kant would have regarded as synthetic a priori, but whose conjunction is incompatible with the principles of special relativity and general covariance. (These include normal induction, the continuity of space, and the euclidean character of space.) A Kantian can consistently hold on to euclidean geometry come what may; but then experience may force him to give up normal induction or the continuity of space. Or he may hold on to normal induction and the continuity of space come what may; but then experience may force him to give up euclidean geometry (this happens in the case that physical space is not even homeomorphic to any euclidean space). In his article in *Albert Einstein, Philosopher-Scientist*, Reichenbach gives essentially the same argument in a slightly different form.

Applied to our present context, what this shows is that there are principles such that philosophers fond of the overblown notion of analyticity, and in particular philosophers who identify analyticity with (maximum) unrevisability, would classify them as analytic, but whose conjunction has testable empirical consequences. Thus either the identification of analyticity with centrality must be given up once and for all, or one must give up the idea that analyticity is closed under conjunction, or one must swallow the unhappy consequence that an analytic sentence can have testable empirical consequences (and hence that an *analytic* sentence might turn out to be *empirically false*).

It is no accident, by the way, that the sentences that Kant would have classified as synthetic a priori would be classified by these latter-day empiricists as analytic; their purpose in bloating the notion of analyticity was precisely to dissolve Kant's problem by identifying a pri-

oricity with analyticity and then identifying analyticity in turn with truth by convention. (This last step has also been devastatingly criticized by Quine, but discussion of it would take us away from our topic.)

Other philosophers have tried to answer Quine by distinguishing between *sentences* and *statements*: all *sentences* are revisable, they agree, but some *statements* are not. Revising a sentence is not changing our mind about the statement formerly expressed by that sentence just in case the sentence (meaning the syntactical object together with its meaning) after the revision is, in fact, not synonymous with the sentence prior to the revision, i.e., just in case the revision is a case of meaning change and not change of theory. But (1) this reduces at once to the proposal to explicate analyticity in terms of synonymy; and (2) if there is one thing that Quine has decisively contributed to philosophy, it is the realization that meaning change and theory change cannot be sharply separated. We do not agree with Quine that meaning change cannot be defined at all, but it does not follow that the dichotomy "meaning change or theory change" is tenable. Discovering that we live in a non-euclidean world *might* change the meaning of "straight line" (this would happen in the — somewhat unlikely — event that something like the parallels postulate was part of the stereotype of straightness); but it would not be a *mere* change of meaning. In particular it would not be a change of *extension*: thus it would not be right to say that the parallels postulate was "true in the former sense of the words." From the fact that giving up a sentence *S* would involve meaning change, it does not follow that *S* is *true*. Meanings may not fit the world; and meaning change can be forced by empirical discoveries.

Although we are not, in this paper, trying to explicate a notion of analyticity, we are trying to explicate a notion that might seem closely related, the notion of meaning. Thus it might seem that Quine's arguments would also go against our attempt. Let us check this out.

On our view there is a perfectly good sense in which being striped is part of the meaning of "tiger." But it does not follow, on our view, the "tigers are striped" is analytic. If a mutation occurred, all tigers might be albinos. Communication presupposes that I have a stereotype of tigers which includes stripes, and that you have a stereotype of tigers which includes stripes, and that I know that your stereotype includes stripes, and that you know that my stereotype includes stripes, and that you

know that I know . . . (and so on, a la Grice, forever). But it does not presuppose that any particular stereotype be *correct*, or that the majority of our stereotypes remain correct forever. Linguistic obligatoriness is not supposed to be an index of unrevisability or even of truth; thus we can hold that "tigers are striped" is part of the meaning of "tiger" without being trapped in the problems of analyticity.

Thus Quine's arguments against identifying analyticity with centrality are not arguments against identifying a feature's being "part of the meaning" of *X* with its being obligatorily included in the stereotype of *X*. What of Quine's "noise" argument?

Of course, evidence concerning what people say, including explicit metalinguistic remarks, is important in "semantics" as it is in syntax. Thus, if a speaker points to a *clam* and asks "is that a tiger?" people are likely to guffaw. (When they stop laughing) they might say "he doesn't know the meaning of 'tiger,'" or "he doesn't know what tigers are." Such comments can be helpful to the linguist. But we are not *defining* the stereotype in terms of such comments. To say that being "big-cat-like" is part of the meaning of tiger is not merely to say that application of "tiger" to something which is not big-cat-like (and also not a tiger) would provoke certain *noises*. It is to say that speakers acquire the information that "tigers are (stereotypically) big-cat-like" as they acquire the word "tiger" and that they feel an obligation to guarantee that those to whom they teach the use of the word do likewise. Information about the minimum skills required for entry into the linguistic community is significant information; no circularity of the kind Quine criticized appears here.

Radical translation. What our theory does not do, by itself at any rate, is solve Quine's problem of "radical translation" (i.e., translation from an alien language/culture). We cannot translate our hypothetical Cheroquoi into English by matching stereotypes, just because finding out what the stereotype of, say, *wa'arabi* is involves translating Cheroquoi utterances. On the other hand, the constraint that each word in Cheroquoi should match its image in English under the translation-function as far as stereotype is concerned (or approximately match, since in many cases exact matching may not be attainable), places a severe *constraint* on the translation-function. Once we have succeeded in translating the basic vocabulary of Cheroquoi, we can start to elicit stereotypes, and these will serve both to constrain future translations and to check the

internal correctness of the piece of the translation-function already constructed.

Even where we can determine stereotypes (relative, say, to a tentative translation of "basic vocabulary"), these do not suffice, in general, to determine a unique translation. Thus the German words *Ulme* and *Buche* have the same stereotype as *elm*; but *Ulme* means "elm" while *Buche* means "beech." In the case of German, the fact that *Ulme* and *elm* are cognates could point to the correct translation (although this is far from foolproof — in general, cognate words are not synonymous); but in the case of Greek we have no such clue as to which of the two words *'οξύα, πτελέα* means *elm* and which *beech*; we would just have to find a Greek who could tell elms from beeches (or oxya from *ptelea*). What this illustrates is that it may not be the *typical* speakers' dispositions to assent and dissent that the linguist must seek to discover; because of the division of linguistic labor, it is frequently necessary for the linguist to assess who are the experts with respect to oxya, or *wa'arabi*, or *gavagai*, or whatever, before he can make a guess at the socially determined extension of a word. Then this socially determined extension *and* the stereotype of the *typical* speaker, inexpert though he is, will *both* function as constraints upon the translation-function. Discovery that the stereotype of oxya is wildly different from the stereotype of *elm* would disqualify the translation of oxya by *elm* in all save the most extensional contexts; but the discovery that the *extension* of oxya is not even approximately the class of elms would wipe out the translation altogether, in all contexts.

It will be noted that we have already enlarged the totality of facts counted as evidence for a translation-function beyond the ascetic base that Quine allows in *Word and Object*. For example, the fact that speakers say such-and-such when the linguist's "confederate" points to the word oxya and asks "what does this mean?" or "what is this?" or whatever is not allowed by Quine (as something the linguist can "know") on the ground that this sort of "knowledge" presupposes already having translated the query "what does this word mean?". However, if Quine is willing to assume that one can *somehow* guess at the words which signify *assent* and *dissent* in the alien language, it does not seem at all unreasonable to suppose that one can somehow convey to a native speaker that one does not understand a word. It is not necessary that one discover a locution in the alien language which literally means

"what does this word *mean*?" (as opposed to: "I don't understand this word," or "this word is unfamiliar to me" or "I am puzzled by this word," etc.). Perhaps just saying the word oxya, or whatever, with a tone of puzzlement would suffice. Why should *puzzlement* be less accessible to the linguist than *assent*?

Also, we are taking advantage of the fact that segmentation into *words* has turned out to be linguistically universal (and there even exist tests for word and morpheme segmentation which are independent of meaning). Clearly, there is no motivated reason for allowing the linguist to utter whole sentences and look for assent and dissent, while refusing to allow him to utter words and morphemes in a tone of puzzlement.

I repeat, the claim is not being advanced that enlarging the evidence base in this way solves the problem of radical translation. What it does is add further constraints on the class of admissible candidates for a correct translation. What I believe is that enlarging the class of constraints can determine a unique translation, or as unique a translation as we are able to get in practice. But constraints that go beyond linguistic theory proper will have to be used, in my opinion; there will also have to be constraints on what sorts of beliefs (and connections between beliefs, and connections of beliefs to the culture and the world) we can reasonably impute to people. Discussion of these matters will be deferred to another paper.

A critique of Davidsonian semantic theory. In a series of publications, Donald Davidson has put forward the interesting suggestion that a semantic theory of a natural language might be modeled on what mathematical logicians call a *truth definition* for a formalized language. Stripped of technicalities, what this suggestion comes down to is that one might have a set of rules specifying (1) for each word, under what conditions that word is true of something (for words for which the concept of an extension makes sense; all other words are to be treated as syncategorematic); (2) for sentences longer than a single word, a rule is given specifying the conditions under which the sentence is true as a function of the way it is built up out of shorter sentences (counting words as if they were one-word sentences, e.g., "snow" as "that's snow"). The choice of one-word sentences as the starting point is my interpretation of what Davidson intends; in any case, he means one to start with a *finite* stock of *short* sentences for which truth conditions are to be laid

down *directly*. The intention of (2) is not that there should be a rule for each sentence not handled under (1), since this would require an infinite number of rules, but that there should be a rule for each sentence *type*. For example, in a formalized language one of the rules of kind (2) might be: if S is $(S_1 \,\&\, S_2)$ for some sentences S_1,S_2, then S is true if and only if S_1,S_2, are both true.

It will be noticed that, in the example just given, the truth condition specified for sentences of the sentence type $(S_1 \,\&\, S_2)$ performs the job of specifying the meaning of "&." More precisely, it specifies the meaning of the structure (—— & ——). This is the sense in which a truth definition can be a theory of meaning. Davidson's contention is that the *entire* theory of meaning for a natural language can be given in this form.

There is no doubt that rules of the type illustrated can give the meaning of some words and structures. The question is, what reason is there to think that the meaning of most words can be given in this way, let alone all?

The obvious difficulty is this: for many words, an extensionally correct truth definition can be given which is in no sense a theory of the meaning of the word. For example, consider *"Water" is true of x if and only if x is H_2O*. This is an extensionally correct truth definition for "water" (strictly speaking, it is not a truth definition but a "truth of" definition — i.e., a *satisfaction*-in-the-sense-of-Tarski definition, but we will not bother with such niceties here). At least it is extensionally correct if we ignore the problem that water with impurities is also called "water," etc. Now, suppose most speakers don't *know* that water is H_2O. Then this formula in no way tells us anything about the *meaning* of "water." It might be of interest to a chemist, but it doesn't count as a theory of the meaning of the term "water." Or, it counts as a theory of the *extension* of the term "water," but Davidson is promising us more than just that.

Davidson is quite well aware of this difficulty. His answer (in conversation, anyway) is that we need to develop a theory of *translation*. This he, like Quine, considers to be the real problem. Relativized to such a theory (relativized to what we admittedly don't yet have), the theory comes down to this: we want a system of truth definitions which is simultaneously a system of translations (or approximate translations, if perfect translation is unobtainable). If we had a theory which specified what it is to be a good translation, then we could rule out the above truth definition for "water" as uninteresting on the grounds that x is H_2O is not an acceptable translation or even near-translation of *x is water* (in a prescientific community), even if water = H_2O happens to be true.

This comes perilously close to saying that a theory of meaning is a truth definition plus a theory of meaning. (If we had ham and eggs we'd have ham and eggs — *if* we had ham and *if* we had eggs.) But this story suffers from worse than promissoriness, as we shall see.

A second contention of Davidson's is that the theory of translation that we don't yet have is necessarily a theory whose basic units are *sentences* and not *words* on the grounds that our *evidence* in linguistics necessarily consists of assent and dissent from sentences. Words can be handled, Davidson contends, by treating them as sentences ("water" as "that's water," etc.).

How does this ambitious project of constructing a theory of meaning in the form of a truth definition constrained by a theory of translation tested by "the only evidence we have," speakers' dispositions to use sentences, fare according to the view we are putting forward here?

Our answer is that the theory cannot succeed in principle. In special cases, such as the word "and" in its truth-functional sense, a truth definition (strictly speaking, a clause in what logicians call a "truth definition" — the sum total of all the clauses is the inductive definition of "truth" for the particular language) can give the meaning of the word or structure because the stereotype associated with the word (if one wants to speak of a stereotype in the case of a word like "and") is so strong as to actually constitute a necessary and sufficient condition. If all words were like "and" and "bachelor" the program could succeed. And Davidson certainly made an important contribution in pointing out that linguistics has to deal with inductively specified truth conditions. But for the great majority of words, the requirements of a theory of truth and the requirements of a theory of meaning are mutually incompatible, at least in the English-English case. But the English-English case — the case in which we try to provide a significant theory of the meaning of English words which is itself couched in English — is surely the basic one.

The problem is that in general the only expressions which are both coextensive with X and have roughly the same stereotype as X are ex-

pressions containing X itself. If we rule out such truth definitions (strictly speaking, clauses, but I shall continue using "truth definition" both for individual clauses and for the whole system of clauses, for simplicity) as

"X is water" is true if and only if X is water

on the grounds that they don't say anything about the meaning of the word "water," and we rule out such truth definitions as

"X is water" is true if and only if X is H_2O

on the grounds that what they say is wrong as a description of the *meaning* of the word "water," then we shall be left with nothing.

The problem is that we want

W is true of x if and only if ——

to satisfy the conditions that (1) the clause be extensionally correct (where —— is to be thought of as a condition containing "x," e.g. "x is H_2O"); (2) that —— be a *translation* of W — on our theory, this would mean that the stereotype associated with W is approximately the same as the stereotype associated with ——; (3) that —— not contain W itself, or syntactic variants of W. If we take W to be, for example, the word "elm," then there is absolutely no way to fulfill all three conditions simultaneously. Any condition of the above form that does not contain "elm" and that is extensionally correct will contain a —— that is absolutely terrible as a *translation* of "elm."

Even where the language contains two exact synonyms, the situation is little better. Thus

"Heather" is true of x if and only if x is gorse

is true, and so is

"Gorse" is true of x if and only if x is heather

—— *this* is a *theory* of the *meaning* of "gorse" and "heather"?

Notice that the condition (3) is precisely what logicians do *not* impose on *their* truth definitions.

"Snow is white" is true if and only if snow is white

is the paradigm of a truth definition in the logician's sense. But logicians are trying to give the extension of "true" with respect to a particular language, not the meaning of "snow is white." Tarski would

have gone so far as to claim he was giving the *meaning* (and not just the extension) of "true"; but he would never have claimed he was saying *anything* about the meaning of "snow is white."

It may be that what Davidson really thinks is that theory of meaning, in any serious sense of the term, is impossible, and that all that is possible is to construct translation-functions. If so, he might well think that the only "theory of meaning" possible for English is one that says " 'elm' is true of x if and only if x is an elm," " 'water' is true of x if and only if x is water," etc., and only rarely something enlightening like "S_1 & S_2 is true if and only if S_1,S_2 are both true." But if Davidson's "theory" is just Quinine skepticism under the disguise of a positive contribution to the study of meaning, then it is a bitter pill to swallow.

The contention that the only evidence available to the linguist is speakers' dispositions with respect to whole sentences is, furthermore, vacuous on one interpretation, and plainly false on the interpretation on which it is not vacuous. If dispositions to say certain things *when queried about individual words or morphemes or syntactic structures* are included in the notion of dispositions to use sentences, then the restriction to dispositions to use sentences seems to rule out nothing whatsoever. On the non-vacuous interpretation, what Davidson is saying is that the linguist cannot have access to such data as what informants (including the linguist himself) say when asked the meaning of a word or morpheme or syntactic structure. No reason has ever been given why the linguist cannot have access to such data, and it is plain that actual linguists place heavy reliance on informants' testimony about such matters, in the case of an alien language, and upon their own intuitions as native speakers, when they are studying their native languages. In particular, when we are trying to translate a whole sentence, there is no reason why we should not be guided by our knowledge of the syntactic and semantic properties of the constituents of that sentence, including the deep structure. As we have seen, there are procedures for gaining information about individual constituents. It is noteworthy that the procedure that Quine and Davidson claim is the only *possible* one — going from whole sentences to individual words — is the *opposite* of the procedure upon which every success ever attained in the study of natural language has been based.

Critique of California semantics. I wish now to consider an approach to semantic theory pioneered by the late Rudolf Carnap. Since I do not

wish to be embroiled in textual questions, I will not attribute the particular form of the view I am going to describe to any particular philosopher but will simply refer to it as "California semantics."

We assume the notion of a *possible world*. Let f be a function defined on the "space" of all possible worlds whose value $f(x)$ at any possible world x is always a subset of the set of entities in x. Then f is called an *intension*. A term T has meaning for a speaker X if X associates T with an intension f_T. The term T is *true of* an entity e in a possible world x if and only if e belongs to the set $f(x)$. Instead of using the term "associate," Carnap himself tended to speak of "grasping" intensions; but, clearly, what was intended was not just that X "grasp" the intension f, but that he grasp *that* f is the intension *of* T — i.e., that he *associate* f with T in some way.

Clearly this picture of what it is to understand a term disagrees with the story we tell in this paper. The reply of a California semanticist would be that California semantics is a description of an *ideal* language; that actual language is *vague*. In other words, a term T in actual language does not have a single precise intension; it has a set — possibly a fuzzy set — of intensions. Nevertheless, the first step in the direction of describing natural language is surely to study the idealization in which each term T has exactly one intension.

(In his book *Meaning and Necessity*, Carnap employs a superficially different formulation: an intension is simply a *property*. An entity e belongs to the extension of a term T just in case e has whichever property is the intension of T. The later formulation in terms of functions f as described above avoids taking the notion of *property* as primitive.)

The first difficulty with this position is the use of the totally unexplained notion of *grasping* an intension (or, in our reformulation of the position, *associating* an intension with a term). Identifying intensions with set-theoretic entities f provides a "concrete" realization of the notion of intension in the current mathematical style (relative to the notions of possible world and set), but at the cost of making it very difficult to see how anyone could have an intension in his mind, or what it is to think about one or "grasp" one or "associate" one with anything. It will not do to say that thinking of an intension is using a word or functional substitute for a word (i.e., the analogue of a word in "brain code," if, as seems likely, the brain "computes" in a "code" that has analogies to and possibly borrowings from language; or a thought

form such as a picture or a private symbol, in cases where such are employed in thinking) which *refers* to the intension in question, since *reference* (i.e., being in the extension of a term) has just been defined in terms of *intension*. Although the characterization of what it is to think of an abstract entity such as a function or a property is certainly correct, in the present context it is patently circular. But no non-circular characterization of this fundamental notion of the theory has ever been provided.

This difficulty is related to a general difficulty in the philosophy of mathematics pointed out by Paul Benacerraf.[24] Benacerraf has remarked that philosophies of mathematics tend to fall between two stools: either they account for what mathematical objects are and for the necessity of mathematical truth and fail to account for the fact that people can *learn* mathematics, can *refer to* mathematical objects, etc., or else they account for the latter facts and fail to account for the former. California semantics accounts for what intensions are, but provides no account that is not completely circular of how it is that we can "grasp" them, associate them with terms, think about them, *refer to* them, etc.

Carnap may not have noticed this difficulty because of his Verificationism. In his early years Carnap thought of understanding a term as possessing the *ability to verify* whether or not any given entity falls in the extension of the term. In terms of intensions: "grasping" an intension would amount, then, to possessing the ability to verify if an entity e in any possible world x belongs to $f(x)$ or not. Later Carnap modified this view, recognizing that, as Quine puts it, sentences face the tribunal of experience collectively and not individually. There is no such thing as the way of verifying that a term T is true of an entity, in general, independent of the context of a particular set of theories, auxiliary hypotheses, etc. Perhaps Carnap would have maintained that something like the earlier theory was correct for a limited class of terms, the so-called "observation terms." Our own view is that the verifiability theory of meaning is false both in its central idea and for observation terms, but we shall not try to discuss this here. At any rate, if one is *not* a verificationist, then it is hard to see California semantics as a theory at all, since the notion of *grasping* an intension has been left totally unexplained.

Second, if we assume that "grasping an intension" (associating an in-

[24] See his "Mathematical Truth," *Journal of Philosophy*, 70 (1973): 661–678.

tension with a term T) is supposed to be a *psychological state* (in the narrow sense), then California semantics is committed to both principles (1) and (2) that we criticized in the first part of this paper. It must hold that the psychological state of the speaker determines the intension of his terms which in turn determines the extension of his terms. It would follow that if two human beings are in the same total psychological state, then they necessarily assign the same extension to every term they employ. As we have seen, this is totally wrong for natural language. The reason this is wrong, as we saw above, is in part that extension is determined socially, not by individual competence alone. Thus California semantics is committed to treating language as something private — to totally ignoring the linguistic division of labor. The extension of each term is viewed by this school as totally determined by something in the head of the individual speaker all by himself. A second reason this is wrong, as we also saw, is that most terms are *rigid*. In California semantics every term is treated as, in effect, a *description*. The *indexical* component in meaning — the fact that our terms refer to things which are similar, in certain ways, to things that we designate *rigidly*, to *these* things, to the stuff we call "water," or whatever, *here* — is ignored.

But what of the defense that it is not actual language that the California semanticist is concerned with, but an idealization in which we "ignore vagueness," and that terms in natural language may be thought of as associated with a set of intensions rather than with a single well-defined intension?

The answer is that an *indexical* word cannot be represented as a vague family of non-indexical words. The word "I," to take the extreme case, is *indexical* but not *vague*. "I" is not synonymous with a *description*; neither is it synonymous with a fuzzy set of descriptions. Similarly, if we are right, "water" is synonymous neither with a description nor with a fuzzy set of descriptions (intensions).

Similarly, a word whose extension is fixed socially and not individually is not the same thing as a word whose extension is *vaguely* fixed individually. The reason my individual "grasp" of "elm tree" does not fix the extension of elm is not that the word is vague — if the problem were simple vagueness, then the fact that my concepts do not distinguish elms from beeches would imply that elms are beeches, as I use the term, or, anyway, borderline cases of beeches, and that beeches are elms, or borderline cases of elms. The reason is rather that the extension of "elm

tree" in my dialect is not fixed by what the average speaker "grasps" or doesn't "grasp" at all; it is fixed by the community, including the experts, through a complex cooperative process. A language which exemplifies the division of linguistic labor cannot be approximated successfully by a language which has vague terms and no linguistic division of labor. Cooperation isn't vagueness.

But, one might reply, couldn't one replace our actual language by a language in which (1) terms were replaced by coextensive terms which were *not* indexical (e.g., "water" by "H_2O," assuming "H_2O" is not indexical); and (2) we eliminated the division of linguistic labor by making every speaker an expert on every topic?

We shall answer this question in the negative; but suppose, for a moment, the answer were "yes." What significance would this have? The "ideal" language would in no sense be similar to our actual language; nor would the difference be a matter of "the vagueness of natural language."

In fact, however, one can't carry out the replacement, for the very good reason that *all* natural-kind words and physical-magnitude words are indexical in the way we have described, "hydrogen," and hence "H_2O," just as much as "water." Perhaps "sense data" terms are not indexical (apart from terms for the self), if such there be; but "yellow" as a *thing* predicate is indexical for the same reason as "tiger"; even if something *looks* yellow it may not *be* yellow. And it doesn't help to say that things that look yellow in normal circumstances (to normal perceivers) are yellow; "normal" here has precisely the feature we called indexicality. There is simply no reason to believe that the project of reducing our language to non-indexical language could be carried out in principle.

The elimination of the division of linguistic labor might, I suppose, be carried out "in principle." But, if the division of linguistic labor is, as I conjectured, a linguistic universal, what interest is there in the possible existence of a language which lacks a constitutive feature of *human* language? A world in which every one is an expert on every topic is a world in which social laws are almost unimaginably different from what they now are. What is the *motivation* for taking such a world and such a language as the model for the analysis of *human* language?

Incidentally, philosophers who work in the tradition of California semantics have recently begun to modify the scheme to overcome just

these defects. Thus it has been suggested that an intension might be a function whose arguments are not just possible worlds but, perhaps, a possible world, a speaker, and a non-linguistic context of utterance. This would permit the representation of some kinds of indexicality and some kinds of division of linguistic labor in the model. As David Lewis develops these ideas, "water," for example, would have the same *intension* (same function) on Earth and on Twin Earth, but a different extension. (In effect, Lewis retains assumption (1) from the discussion in the first part of this paper and gives up (2); we chose to give up (1) and retain (2).) There is no reason why the formal models developed by Carnap and his followers should not prove valuable when so modified. Our interest here has been not in the utility of the mathematical formalism but in the philosophy of language underlying the earlier versions of the view.

Semantic markers. If the approach suggested here is correct, then there is a great deal of scientific work to be done in (1) finding out what sorts of items can appear in stereotypes; (2) working out a convenient system for representing stereotypes; etc. This work is not work that can be done by philosophical discussion, however. It is rather the province of linguistics and psycholinguistics. One idea that can, I believe, be of value is the idea of a *semantic marker*. The idea comes from the work of J. J. Katz and J. A. Fodor; we shall modify it somewhat here.

Consider the stereotype of "tiger" for a moment. This includes such features as being an animal; being big-cat-like; having black stripes on a yellow ground (yellow stripes on a black ground?); etc. Now, there is something very special about the feature *animal*. In terms of Quine's notion of *centrality* or *unrevisability*, it is qualitatively different from the others listed. It is not impossible to imagine that tigers might not be animals (they might be robots). But spelling this out, they must always have been robots; we don't want to tell a story about the tigers being *replaced* by robots, because then the robots wouldn't be tigers. Or, if they weren't always robots, they must have *become* robots, which is even harder to imagine. If tigers are and always were robots, these robots mustn't be too "intelligent," or else we may not have a case in which tigers aren't animals — we may, rather, have described a case in which some robots are animals. Best make them "other directed" robots — say, have an operator on Mars controlling each motion remotely. Spelling this out, I repeat, is difficult, and it is curiously hard to think of the case

to begin with, which is why it is easy to make the mistake of thinking that it is "logically impossible" for a tiger *not* to be an animal. On the other hand, there is no difficulty in imagining an individual tiger that is not striped; it might be an albino. Nor is it difficult to imagine an individual tiger that doesn't look like a big cat: it might be horribly deformed. We can even imagine the whole species losing its stripes or becoming horribly deformed. But tigers ceasing to be animals? Great difficulty again!

Notice that we are not making the mistake that Quine rightly criticized, of attributing an absolute unrevisability to such statements as "tigers are animals," "tigers couldn't change from animals into something else and still be tigers." Indeed, we can describe farfetched cases in which these statements would be given up. But we maintain that it is *qualitatively* harder to revise "all tigers are animals" than "all tigers have stripes" — indeed, the latter statement is not even true.

Not only do such features as "animal," "living thing," "artifact," "day of the week," "period of time," attach with enormous centrality to the words "tiger," "clam," "chair," "Tuesday," "hour"; but they also form part of a widely used and important *system of classification*. The centrality guarantees that items classified under these headings virtually never have to be reclassified; thus these headings are the natural ones to use as category-indicators in a host of contexts. It seems to me reasonable that, just as in syntax we use such markers as "noun," "adjective," and, more narrowly, "concrete noun," "verb taking a person as subject and an abstract object," etc., to classify words, so in semantics these category-indicators should be used as markers.

It is interesting that when Katz and Fodor originally introduced the idea of a semantic marker, they did not propose to exhaust the meaning — what we call the stereotype — by a list of such markers. Rather, the markers were restricted to just the category-indicators of high centrality, which is what we propose. The remaining features were simply listed as a "distinguisher." Their scheme is not easily comparable with ours, because they wanted the semantic markers *plus* the distinguisher to always give a necessary and sufficient condition for membership in the extension of the term. Since the whole thing — markers and distinguisher — were supposed to represent what every speaker implicitly knows, they were committed to the idea that every speaker implicitly knows of a necessary and sufficient condition for membership in the extension of

"gold," "aluminum," "elm" — which, as we have pointed out, is not the case. Later Katz went further and demanded that *all* the features constitute an *analytically* necessary and sufficient condition for membership in the extension. At this point he dropped the distinction between markers and distinguishers; if all the features have, so to speak, the infinite degree of centrality, why call some "markers" and some "distinguishers"? From our point of view, their original distinction between "markers" and "distinguishers" was sound — provided one drop the idea that the distinguisher provides (together with the markers) a necessary and sufficient condition, and the idea that any of this is a theory of *analyticity*. We suggest that the idea of a semantic marker is an important contribution, when taken as suggested here.

The meaning of "meaning." We may now summarize what has been said in the form of a proposal concerning how one might reconstruct the notion of "meaning." Our proposal is not the only one that might be advanced on the basis of these ideas, but it may serve to encapsulate some of the major points. In addition, I feel that it recovers as much of ordinary usage in common sense talk and in linguistics as one is likely to be able to conveniently preserve. Since, on my view something like the assumptions (I) and (II) listed in the first part of this paper are deeply embedded in ordinary meaning talk, and these assumptions are jointly inconsistent with the facts, no reconstruction is going to be without some counter-intuitive consequences.

Briefly, my proposal is to define "meaning" not by picking out an object which will be identified with the meaning (although that might be done in the usual set-theoretic style if one insists), but by specifying a normal form (or, rather, a *type* of normal form) for the description of meaning. If we know what a "normal form description" of the meaning of a word should be, then, as far as I am concerned, we know what meaning *is* in any scientifically interesting sense.

My proposal is that the normal form description of the meaning of a word should be a finite sequence, or "vector," whose components should certainly include the following (it might be desirable to have other types of components as well): (1) the syntactic markers that apply to the word, e.g., "noun"; (2) the semantic markers that apply to the word, e.g., "animal," "period of time"; (3) a description of the additional features of the stereotype, if any; (4) a description of the extension.

The following convention is a part of this proposal: the components of the vector all represent a hypothesis about the individual speaker's competence, *except the extension*. Thus the normal form description for "water" might be, in part:

Syntactic Markers	*Semantic Markers*	*Stereotype*	*Extension*
mass noun, concrete	natural kind, liquid	colorless, transparent, tasteless, thirst-quenching, etc.	H_2O (give or take impurities)

— this does *not* mean that knowledge of the fact that water is H_2O is being imputed to the individual speaker or even to the society. It means that (*we* say) the extension of the term "water" as *they* (the speakers in question) use it is *in fact* H_2O. The objection "who are we to say what the extension of *their* term is in fact" has been discussed above. Note that this is fundamentally an objection to the notion of *truth*, and that extension is a relative of truth and inherits the family problems.

Let us call two descriptions *equivalent* if they are the same except for the description of the extension, and the two descriptions are coextensive. Then, if the set variously described in the two descriptions is, *in fact*, the extension of the word in question, and the other components in the description are correct characterizations of the various aspects of competence they represent, *both* descriptions count as correct. Equivalent descriptions are both correct or both incorrect. This is another way of making the point that, although we have to use a *description* of the extension to *give* the extension, we think of the component in question as being the *extension* (the *set*), not the description of the extension.

In particular the representations of the words "water" in Earth dialect and "water" in Twin Earth dialect would be the same except that in the last column the normal form description of the Twin Earth word "water" would have XYZ and not H_2O. This means, in view of what has just been said, that we are ascribing the *same* linguistic competence to the typical Earthian/Twin Earthian speaker, but a different extension to the word, nonetheless.

This proposal means that we keep assumption (II) of our early dis-

cussion. Meaning determines extension — by construction, so to speak. But (I) is given up; the psychological state of the individual speaker does not determine "what he means."

In most contexts this will agree with the way we speak, I believe. But one paradox: suppose Oscar is a German-English bilingual. On our view, in his total collection of dialects, the words *beech* and *Buche* are exact synonyms. The normal form descriptions of their meanings would be identical. But he might very well not know that they are synonyms! A speaker can have two synonyms in his vocabulary and not know that they are synonyms!

It is instructive to see how the failure of the apparently obvious "if S_1 and S_2 are synonyms and Oscar understands both S_1 and S_2 then Oscar knows that S_1 and S_2 are synonyms" is related to the falsity of (I), on our analysis. Notice that if we had chosen to omit the extension as a component of the "meaning-vector," which is David Lewis's proposal as I understand it, then we would have the paradox that "elm" and "beech" have the *same meaning* but different extensions!

On just about any materialist theory, believing a proposition is likely to involve processing some *representation* of that proposition, be it a sentence in a language, a piece of "brain code," a thought form, or whatever. Materialists, and not only materialists, are reluctant to think that one can believe propositions *neat*. But even materialists tend to believe that, if one believes a proposition, *which* representation one employs is (pardon the pun) immaterial. If S_1 and S_2 are both representations that are *available* to me, then if I believe the proposition expressed by S_1 under the representation S_1, I must also believe it under the representation S_2 — at least, I must do this if I have any claim to rationality. But, as we have just seen, this isn't right. Oscar may well believe that *this* is a "beech" (it has a sign on it that says "beech"), but not believe or disbelieve that this is a "*Buche*." It is not just that belief is a process involving representations; he believes the proposition (if one wants to introduce "propositions" at all) under one representation and not under another.

The amazing thing about the theory of meaning is how long the subject has been in the grip of philosophical misconceptions, and how strong these misconceptions are. Meaning has been identified with a necessary and sufficient condition by philosopher after philosopher. In the empiricist tradition, it has been identified with method of verifica-

tion, again by philosopher after philosopher. Nor have these misconceptions had the virtue of exclusiveness; not a few philosophers have held that meaning = method of verification = necessary and sufficient condition.

On the other side, it is amazing how weak the grip of the facts has been. After all, what have been pointed out in this essay are little more than home truths about the way we use words and how much (or rather, how little) we actually know when we use them. My own reflection on these matters began after I published a paper in which I confidently maintained that the meaning of a word was "a battery of semantical rules,"[25] and then began to wonder how the meaning of the common word "gold" could be accounted for in this way. And it is not that philosophers had never considered such examples: Locke, for example, uses this word as an example and is not troubled by the idea that its meaning is a necessary and sufficient condition!

If there is a reason for both learned and lay opinion having gone so far astray with respect to a topic which deals, after all, with matters which are in everyone's experience, matters concerning which we all have more data than we know what to do with, matters concerning which we have, if we shed preconceptions, pretty clear intuitions, it must be connected to the fact that the grotesquely mistaken views of language which are and always have been current reflect two specific and very central philosophical tendencies: the tendency to treat cognition as a purely *individual* matter and the tendency to ignore the *world*, insofar as it consists of more than the individual's "observations." Ignoring the division of linguistic labor is ignoring the social dimension of cognition; ignoring what we have called the *indexicality* of most words is ignoring the contribution of the environment. Traditional philosophy of language, like much traditional philosophy, leaves out other people and the world; a better philosophy and a better science of language must encompass both.

[25] "How Not to Talk About Meaning," in R. Cohen and M. Wartofsky, eds., *Boston Studies in the Philosophy of Science*, vol. 2 (New York: Humanities Press, 1965).